Neel Gali

Machine Learning Engineer – MLOps & Agentic AI

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Professional Experience

Senior Software Engineer *Jul 2025 – Present*

Epik Solutions | Enterprise AI & MLOps Platform

- Architected and deployed production-grade **LangGraph-based agentic systems** with **LLMs** and custom tool integrations, automating complex business workflows and reducing manual processing time by 30%.
- Built and maintained end-to-end **MLOps pipelines** using **MLflow** for experiment tracking, model versioning, and deployment orchestration on **Databricks**, accelerating model releases by 25%.
- Developed scalable **PySpark** data pipelines on **Databricks** processing 100GB+ daily data, creating optimized UDFs for feature engineering and improving distributed processing efficiency by 35%.
- Designed advanced **RAG architectures** integrating **FAISS**, **Pinecone**, and **Weaviate** vector databases with **LangChain**, enhancing retrieval accuracy by 40% for domain-specific queries.
- Implemented **LLMOps workflows** with automated testing, monitoring, and guardrails for production **LLM deployments** on **AWS SageMaker**, ensuring responsible AI practices and 99.9% uptime.
- Engineered maintainable **object-oriented Python** codebases following SOLID principles and design patterns for ML systems, reducing technical debt by 20% and improving code testability.
- Optimized **LLM inference** through quantization and batching techniques on **AWS EC2** instances, reducing API latency by 40% and cutting infrastructure costs by 15%.

Software Engineer *Jul 2024 – Jun 2025*

Samay Systems LLC | Data Engineering & AI Platform

- Maintained and debugged production **ML pipelines** processing 50M+ daily events using **Apache Airflow** and **PySpark**, achieving 99.5% pipeline reliability through monitoring and alerting.
- Built scalable **data processing workflows** with **PySpark** on **Databricks** and **AWS Glue**, writing efficient UDFs for custom transformations and reducing processing time by 30%.
- Deployed **LLM-powered applications** using **Python**, **Hugging Face Transformers**, and **OpenAI API** on **Azure ML**, improving document classification accuracy by 22%.
- Implemented **vector search systems** with **FAISS** and **Elasticsearch** for semantic retrieval in RAG pipelines, reducing query latency by 200ms for 5000+ requests/minute.
- Designed **object-oriented Python** modules for ML feature stores using factory and strategy patterns, improving code reusability by 35% across multiple ML projects.
- Contributed to **MLOps infrastructure** using **Docker**, **Kubernetes**, and **AWS EKS** for model serving, enabling auto-scaling and high availability for ML endpoints.
- Partnered with Data Analysts and business teams to research operational processes, identifying 5+ automation opportunities that drove $500K in annual efficiency gains.
- Integrated **Snowflake** data warehouse for analytics workloads, building Python connectors for ML feature extraction and supporting bi-directional data flows with vector stores.

Software Engineer *Jul 2021 – Aug 2022*

Zemoso Technologies | Backend & Data Services

- Developed robust backend services in **Python** using **FastAPI** and object-oriented design patterns, supporting 50,000+ active users with sub-200ms API response times.
- Built **ETL pipelines** with **Python** and **Pandas** for customer data processing, implementing modular class structures that reduced maintenance effort by 25%.
- Optimized **PostgreSQL** queries and database schemas for high-traffic applications, improving report generation performance by 30% for critical business operations.
- Implemented **CI/CD pipelines** using **Jenkins** and **Docker** for automated testing and deployment, accelerating release cycles by 25% and ensuring code quality.
- Collaborated with Product Managers and stakeholders across multiple concurrent projects in fast-paced environments, consistently delivering features on time with 95% sprint completion rate.

Skills

Languages: Python, PySpark, SQL
MLOps & AI Frameworks: MLflow, LangGraph, LangChain, Hugging Face Transformers, OpenAI API, PyTorch, Weights & Biases, LangSmith
Data & Compute: Databricks, Apache Spark, Apache Airflow, PySpark UDFs, Snowflake, PostgreSQL
Cloud & Infrastructure: AWS (SageMaker, EC2, EKS, Lambda, S3, Glue), Azure (ML), GCP, Docker, Kubernetes, FAISS, Pinecone, Weaviate, Elasticsearch
Core Concepts: MLOps, LLMOps, Agentic AI Systems, RAG Architectures, Object-Oriented Design Patterns, Production ML Pipelines, Distributed Systems, CI/CD

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India